Songyin **Wu**

Ph.D. Student · Computer Graphics

University of California, Santa Barbara, USA

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Education

University of California, Santa Barbara — *California, USA*

Ph.D. student in Computer Science — *Sep. 2022 - Present*

- Advised by **Prof. Ling-Qi Yan**.

Peking University — *Beijing, China*

B.S. in Computer Science (Turing Class) — *Sep. 2017 - Jun. 2021*

- Advised by **Prof. Baoquan Chen**.

Publications

Monocular Online Reconstruction with Enhanced Detail Preservation

ACM SIGGRAPH 2025 (Conference Track)

Songyin Wu, *Zhaoyang Lv, Yufeng Zhu, Duncan Frost, Zhengqin Li, Ling-Qi Yan, Carl Ren, Richard Newcombe, Zhao Dong*

GFFE: G-buffer Free Frame Extrapolation for Low-latency Real-time Rendering

ACM Transactions on Graphics (Proceedings of SIGGRAPH Asia 2024)

Songyin Wu, *Deepak Vembar, Anton Sochenov, Selvakumar Panneer, Sungye Kim, Anton Kaplanyan, Ling-Qi Yan*

ExtraSS: A Framework for Joint Spatial Super Sampling and Frame Extrapolation

ACM SIGGRAPH Asia 2023 (Conference Track)

Songyin Wu, *Sungye Kim, Zheng Zeng, Deepak Vembar, Sangeeta Jha, Anton Kaplanyan, Ling-Qi Yan*

Unified Gaussian Primitives for Scene Representation and Rendering

Preprint

Yang Zhou, ***Songyin Wu****, Ling-Qi Yan*

BSDF Importance Baking: A Lightweight Neural Solution to Importance Sampling General Parametric BSDFs

Preprint

Yaoyi Bai, ***Songyin Wu****, Zheng Zeng, Beibei Wang, Ling-Qi Yan*

Professional Experience

Meta Reality Labs Research — *Washington, USA*

Research Scientist Intern — *Jun. 2024 - Jan. 2025*

- Research project on interactive scene reconstruction.

Intel Corporation — *California, USA*

Research Scientist Intern — *Jun. 2023 - Sep. 2023*

- Research project on the generative model for general rendering tasks.

Microsoft Research Asia — *Beijing, China*

Research Scientist Intern — *Dec. 2021 - Jul. 2022*

- Research project on face re-animation and view synthesis.

Cornell Tech — *Remote*

Research Assistant Intern — *Jun. 2020 - MAy. 2021*

- Research project on image-based relighting and blending.

Skills

Programming	C/C++, Python, LaTex
Software & Tool	PyTorch, Unreal, Falcor, Blender, OpenCV
Languages	Chinese, English